

VINOD KUMAR NELANAKULA -SOFTWARE ENGINEER

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SUMMARY

Full-stack Software Engineer with 3 years of experience building **React/TypeScript** front ends and **Node/Python** backends. Production experience with MongoDB, Redis, BullMQ, Firebase Auth, REST APIs, and real-time features (RTK Query, WebSockets, Supabase Realtime). Built ExecLens with a Node.js instrumentation runner and AST pipeline. Delivered Carrier Support Portal end-to-end and backend POCs for fault detection.

TECHNICAL SKILLS

Backend: Node.js, Express.js, Python, REST APIs, BullMQ, JWT, gRPC (NestJS)

Frontend: React, TypeScript, Redux Toolkit, RTK Query, Next.js

Database & Cache: MongoDB, Redis, SQL, Prisma

Runtime & AST: Acorn, esbuild, Node.js sandbox instrumentation

Auth & Cloud: Firebase Auth, Vercel, Supabase Realtime

Foundations: Data Structures & Algorithms (actively practising)

Tools: Git, Nx, pnpm, Docker, Jira, CircleCI | Testing: Jest, React Testing Library

EXPERIENCE

Software Engineer • Carrier Global Technologies — Apr 2023 – Present | Hyderabad, India

Carrier Support Portal — Full Stack Web App

- Built a full-stack support app: React 18, TypeScript, Redux Toolkit, Firebase Auth on the front end.
- Implemented real-time technician tracking with RTK Query and browser geolocation.
- Integrated MongoDB geo queries for location matching, Redis caching, and BullMQ job queues.
- Added i18n and responsive UI with Material UI and Tailwind CSS.

Live: [customer-support-007.web.app](#)

Carrier Service Tech App — Proof of Concept

- Built a Python backend with MongoDB for HVAC fault detection and diagnostic logic.
- Created question logic engine and RESTful APIs with a modular, expandable structure.
- Delivered self-contained backend POC ready for integration with client applications.

Consumer Portal Web

- Bootstrapped React Native Web app with Webpack/Vite; integrated analytics and auth flows.
- Increased test coverage to 90% with Jest and React Testing Library.

Live: [my.carrier.com](#)

SmartHome App (React Native)

- Integrated real-time air quality APIs and device connectivity workflows.
- Contributed to **React Native platform upgrade**, improving application stability and performance.
- Enhanced reliability by handling real-world edge cases (Wi-Fi connectivity, scan interruptions).
- Delivered features across **authentication**, **device connectivity**, and energy management workflows.
- Improved accessibility with TalkBack support for inclusive mobile UX.
- Implemented localisation and unit handling for global consistency.
- Maintained **production stability** through iterative fixes, reverts, and follow-up releases.

PERSONAL PROJECTS

ExecLens — JavaScript Execution Engine & API

- Built POST /api/run backend that instruments and executes JS/TS snippets in a Node.js sandbox.
- Implemented step recorder for call stack, event loop phases, Promises, async/await, scope, and closures.
- AST pipeline with Acorn + esbuild source maps; virtual clock for setTimeout; Dockerized runner.

Live: js-exec-lens.vercel.app • Code: github.com/Nvinod007/exec-lens

NestJS gRPC Auth Service — Backend Microservice

- Authentication microservice built with NestJS and gRPC for secure, scalable service-to-service auth.
- Code: github.com/Nvinod007/nestjs-grpc-auth-service

Sprint Poker — Supabase Realtime Backend

- Serverless real-time voting rooms with Supabase; no-auth flow for agile estimation sessions.

Live: sprintpoker.vercel.app • Code: github.com/Nvinod007/sprint-poker

DrawnGuess — Multiplayer Game (In Progress)

- Shipped lobby flow: create/join rooms; building socket-based drawing and guessing gameplay.

Live: drawnplay.vercel.app • Code: github.com/Nvinod007/drawnguess

EDUCATION

B.Tech — Computer Science & Engineering, RGUKT IIIT RKV | CGPA: 8.4 | May 2023

CERTIFICATIONS

- Certified SAFe Agile Scrum Master — Scaled Agile
- AWS Certified AI Practitioner — Amazon Web Services (AWS), Apr 2026
- Supervised Machine Learning: Regression and Classification — DeepLearning.AI, Apr 2026